

WAP to display in reverse of numbers from 1 to 10 using List Comprehension
output:
[10, 9, 8, 7, 6, 5, 4, 3, 2, 1]

Your Code

```
1 print([i for i in range(10, 0, -1)])
```

Input

stdin input

Output

```
[10, 9, 8, 7, 6, 5, 4, 3, 2, 1]
```

WAP to find the Sum of First 10 Natural Numbers
output:
55

Your Code

```
1 print(sum([i for i in range(1, 11)]))
```

Input

stdin input

Output

```
55
```

Visible Test Cases

Test Case 1



10.00 / 10 marks

WAP to find the Sum of Odd Numbers up to 10

output:

25

Your Code

```
1 print(sum([x for x in range(1, 11) if x % 2 != 0]))
```

Input

stdin input

Output

25

Visible Test Cases

Test Case 1



WAP to find the Sum of Squares up to 5

output:

55

Your Code

```
1 print(sum([x**2 for x in range(1, 6)]))
```

Input

stdin input

Output

55

Visible Test Cases

Test Case 1



WAP to find the Sum of first 12 Numbers Divisible by 3

Output:

30

Your Code

```
1 print(sum([x for x in range(1, 13) if x % 3 == 0]))
```

Input

stdin input

Output

30

Visible Test Cases

Test Case 1



WAP to find the factorial of a given number

input:

5

output:

120

Your Code

```
1 from math import prod
2 n = 5
3 print(prod([i for i in range(1,n+1)]))
```

Input

5

Output

120

WAP to find the Sum of Factorials

Series: $1! + 2! + 3! + 4!$

Output:

33

Your Code

```
1 import math
2
3 print(sum([math.factorial(i) for i in range(1, 5)]))
```

Input

stdin input

Output

33

WAP to find the Sum of Digits of a Number

input:

4567

output:

22

Your Code

```
1 num = input()
2 print(sum([int(d) for d in num]))
```

Input

4567

Output

22

WAP to find the Sum of Positive Numbers in a List

input:

[10,-5,20,-8,15,-2]

output:

45

Your Code

```
1 import re
2
3 s = input()
4 lst = [int(x) for x in re.findall(r'-?\d+', s)]
5 print(sum([x for x in lst if x > 0]))
```

Input

[10,-5,20,-8,15,-2]

Output

45

Q10 List Comprehension - Sum of Squares of Even Numbers

10 marks

WAP to Find the Sum of the Squares of Even Numbers up to 10

($2^2 + 4^2 + 6^2 + 8^2 + 10^2$)

OUTPUT:

220

Your Code

```
1 print(sum([x**2 for x in range(2, 11, 2)]))
```

Input

stdin input

Output

220